STUDENT ACADEMIC PROGRESSION SYSTEM

Graphical User Interface & Architectural Layout Report

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1. Real-Time Performance Analytics Dashboard (/)

Features responsive KPI metric cards (Total Enrolled, Active Students, On Leave, Dept Avg CGPA), Chart.js semester-wise SGPA progression curve, activity categories doughnut chart, and live feeds for recent seminars and extracurricular activities.

2. Student Directory & Search Center (/students)

Offers multi-criteria searching by student code, full name, or institutional email, along with admission year and enrollment status filters. Includes modal for registering new students with validation rules.

3. 360-Degree Comprehensive Student Profile (/students/)

Central student portfolio displaying personal information, progressive SGPA vs Cumulative CGPA Chart.js curve, expandable semester cards with subject marks tables, secondary school education records, gap year records, and student seminars/activities.

4. Dynamic Subject Marks & SGPA Recalculation Engine

Interactive modal with real-time letter grade predictor based on obtained vs maximum marks. Automatic database cascade recalculates semester SGPA and updates all subsequent progressive CGPA metrics in the student's academic history.

5. Portfolio: Seminars & Co-Curricular Directory (/portfolio)

Consolidated view of student technical seminars, paper presentations, hackathons, and sports achievements with category and student filters.

6. Google Sheets Synchronization Dashboard (/sync)

Displays real-time webhook status, pending outbox events counter, manual 'Bulk Sync All 7 Tables' trigger, and an immutable audit log of recent synchronization operations.

7. Excel Multi-Tab Relational Exporter (/export/excel)

Automated openpyxl engine that streams a styled .xlsx workbook containing all 7 relational tables with colored headers and auto-adjusted columns.